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Byron C. Wallace

Professor & Director of the BS in Artificial Intelligence, Khoury College of Computer Sciences, Northeastern University

Research interests

I work on natural language processing and machine learning, with applications in healthcare and other high-stakes domains. I am especially interested in model interpretability, analysis, and evaluation.

Education

2012 **Ph.D. Computer Science**, Tufts University, Medford, MA
Advisor Carla E. Brodley
Ph.D. Thesis Machine Learning in Health Informatics: Making Better use of Domain Experts.
Selected as The Runner-Up for the 2013 ACM SIGKDD Doctoral Dissertation Award.

2006 **B.S. Computer Science**, University of Massachusetts at Amherst, Amherst, MA
Minor in Philosophy

Appointments & experience

7/2026 – *present* **Senior Advisor for AI Research at Northeastern**
7/2026 – *present* **Professor**
7/2025 – 7/2026 **Interim Associate Dean for Research for Khoury College of CS**
8/2024 – 7/2025 **Interim Associate Dean of Graduate Programs for Khoury College of CS**
1/2023 – 7/2025 **Sy and Laurie Sternberg Interdisciplinary Associate Professor**
6/2021 – *present* **Associate Professor**
9/2018 – *present* **Director of the BS in Data Science (now AI)**
9/2016 – 6/2021 **Assistant Professor**
Khoury College of Computer Sciences, Northeastern University
Assistant Professor (Adjunct)
Health Services and Public Policy, Brown University

9/2014 – 8/2016 **Assistant Professor**
School of Information, University of Texas at Austin
Assistant Professor (Courtesy)
Department of Computer Science, University of Texas at Austin

6/2012 – 8/2014 **Assistant Professor (Research)**
Health Services and Public Policy, Brown University

Academic honors & professional awards

2025 Our paper, "Learning the Wrong Lessons: Syntactic-Domain Spurious Correlations in Language Models", received a Spotlight designation at NeurIPS 2025 (top 4% of 21575 submissions).

2024 Our paper, "Evaluating the Zero-shot Robustness of Instruction-tuned Language Models", received a Spotlight designation at ICLR 2024 (top 5% of 7262 submissions).

2023 Named the *Sy and Laurie Sternberg Interdisciplinary Associate Professor* at Northeastern.

2022 Our paper, "Evaluating Factuality in Text Simplification", was recognized as an *Outstanding Paper* at ACL 2022.

2021 Our paper, "Understanding Clinical Trial Reports: Extracting Medical Entities and Their Relations"—led by my PhD student Ben Nye—won the *Best Student Paper* award at the AMIA Informatics Summit.

2021 Our paper, "What Would it Take to get Biomedical QA Systems into Practice?" received an *Honorable Mention for Best Paper* at the Workshop on Machine Reading for Question Answering (MRQA; co-located with EMNLP).

2021 Recipient of the *Joel and Ruth Spira Excellence in Teaching Award* for the Khoury College of Computer Sciences at Northeastern.

2019 Invited *Early Career Spotlight* talk at the *International Joint Conference on Artificial Intelligence* (IJCAI) 2019.

2018 Recipient of the annual Early Career Award from the *Society for Research Synthesis Methodology (SRSM)*.

2018 NSF CAREER Award: *Structured Scientific Evidence Extraction: Models and Corpora*.

2018 Acknowledged as a *Top Reviewer* for ACL 2018 (`https://acl2018.org/2018/07/02/top-reviewers/`)

2017 Received the *Distinguished Clinical Research Informatics Paper Award* for "PheKnowCloud: A Tool for Evaluating High-Throughput Phenotype Candidates using Online Medical Literature" at the AMIA Joint Summit (joint work with Jette Henderson, Ryan Bridges, Joyce Ho and Joydeep Ghosh).

2015 Our system won the *Healthcare Data Analytics Challenge* at the IEEE International Conference on Healthcare Informatics (ICHI); joint work with Zhiguo Yu and Todd Johnson.

2014 Inducted as a member of the *Society for Research Synthesis Methodology* (SRSM), an interdisciplinary group of researchers interested in quantitative methods for information synthesis; membership is by invitation only.

2013 Recipient of the *ACM SIGKDD Dissertation Award (Runner-Up)*. Association for Computational Machinery (ACM).

2012 Recipient of the *Outstanding Graduate Researcher at the Doctoral Level* award. Tufts University.

2006 Recipient of the *Outstanding Student in the Area of Systems in Computer Science* award. UMass, Amherst.

2006 Recipient of the *Jonathon Edwards Philosophy Essay Prize*. UMass, Amherst.

2006 Recipient of the *Gerald F. Scanlon Student Employee of the Year* Award. UMass, Amherst.

Publications

Conference publications

These do not include workshop publications, which are listed in a separate section (Workshop & symposium publications), below.

1. Sheridan Feucht, Benno Krojer, Sarah Wang, Henry Abrahamsen, Byron C Wallace, and David Bau. Using OCR Heads to Verbalize Image Semantics. In *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2027.
2. Debjyoti Saharoy, Byron C Wallace, and Javad A. Aslam. Don't Pay Attention, PLANT It: Pretraining Attention via Learning-to-Rank. In *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
3. Sanjana Ramprasad, Pranav Mani, Chenhao Tan, Byron C Wallace, and Elisa Ferracane. Many Benign Errors are Better than A Few Severe Ones: Evaluating Hallucination Severity. In *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS; Evaluation and Datasets Track)*, 2026.
4. Kaijie Mo, Thomas Yang, Chantal Shaib, Qing Yao, William Rudman, Ramez Kouzy, Kanishka Misra, Byron C Wallace, and Junyi Jessy Li. What's in a name? morphological shortcuts by llms in pharmacology. In *Proceedings of Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
5. Chantal Shaib, Tuhin Chakrabarty, Diego Garcia-Olano, and Byron C. Wallace. Measuring ai "slop" in text. In *Proceedings of the Conference on Language Models (COLM)*, 2026.
6. Zihao Yang, Mosh Levy, Yoav Goldberg, and Byron C Wallace. Compared to what? baselines and metrics for counterfactual prompting. In *Proceedings of the Conference on Language Models (COLM)*, 2026.
7. Millicent Li, Alberto Mario Ceballos Arroyo, Giordano Rogers, Naomi Saphra, and Byron C Wallace. Do natural language interpretability methods convey privileged information? In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
8. Hadas Orgad, Fazl Barez, Tal Haklay, Isabelle Lee, Marius Mosbach, Anja Reusch, Naomi Saphra, Byron C Wallace, Sarah Wiegrefe, Eric Wong, Ian Tenney, and Mor Geva. Interpretability can be actionable. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
9. Kaijie Mo, Siddhartha Venkatayogi, Chantal Shaib, Ramez Kouzy, Wei Xu, Byron C Wallace, and Junyi Jessy Li. Faithfulness vs. safety: Evaluating llm behavior under counterfactual medical evidence. In *Proceedings of the Findings of Conference of the Association for Computational Linguistics (ACL)*, 2026.
10. Sebastian Joseph, Lily Chen, Barry Wei, Michael Mackert, Iain J Marshall, Paul Pu Liang, Ramez Kouzy, Byron C Wallace, and Junyi Jessy Li. Decide less, communicate more: On the construct validity of end-to-end fact-checking in medicine. In *Proceedings of the Findings of Conference of the Association for Computational Linguistics (ACL)*, 2026.
11. Hiba Ahsan and Byron C Wallace. Can SAEs reveal and mitigate racial biases of LLMs in healthcare? In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.
12. Chantal Shaib, Venkata S Govindarajan, Joe Barrow, Jiuding Sun, Alexa Siu, Byron C Wallace, and Ani Nenkova. Standardizing the measurement of text diversity: A tool and comparative analysis. In Xuebo Liu and Ayu Purwarianti, editors, *Proceedings of the International Joint Conference on Natural Language Processing and Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP/AACL), System Demonstrations*, pages 36–46, 2025.
13. Sanjana Ramprasad and Byron C Wallace. Do Automatic Factuality Metrics Measure Factuality? A Critical Evaluation. In *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
14. Chantal Shaib, Vinith Menon Suriyakumar, Byron C Wallace, and Marzyeh Ghassemi. Learning the Wrong Lessons: Syntactic-Domain Spurious Correlations in Language Models. In *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
15. Hiba Ahsan, Arnab Sen Sharma, Silvio Amir, David Bau, and Byron C Wallace. Elucidating Mechanisms of Demographic Bias in LLMs for Healthcare. In *Proceedings of the Findings of Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
16. Sheridan Feucht, Eric Todd, Byron C Wallace, and David Bau. The Dual-Route Model of Induction. In *Proceedings of the Conference on Language Modeling (COLM)*, 2025.
17. Somin Wadhwa, Chantal Shaib, Silvio Amir, and Byron C Wallace. Who Taught You That? Tracing Teachers in Model Distillation. In *Proceedings of the Findings of Conference of the Association for Computational Linguistics (ACL)*, 2025.

18. Hye Sun Yun, Karen Y.C. Zhang, Ramez Kouzy, Iain James Marshall, Junyi Jessy Li, and Byron C Wallace. Caught in the Web of Words: Do LLMs Fall for Spin in Medical Literature? In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*, 2025.
19. Jaden Fried Fiotto-Kaufman, Alexander Russell Loftus, Eric Todd, Jannik Brinkmann, Koyena Pal, Dmitrii Troitskii, Michael Ripa, Adam Belfki, Can Rager, Caden Juang, Aaron Mueller, Samuel Marks, Arnab Sen Sharma, Francesca Lucchetti, Nikhil Prakash, Carla E. Brodley, Arjun Guha, Jonathan Bell, Byron C. Wallace, and David Bau. NNsight and NDIF: Democratizing Access to Foundation Model Internals. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
20. Somin Wadwha, Silvio Amir, and Byron C. Wallace. Investigating Mysteries of CoT-Augmented Distillation. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 6071–6086, 2024.
21. Chantal Shaib, Yanai Elaza, Jessy Junyi Li, and Byron C. Wallace. Detection and Measurement of Syntactic Templates in Generated Text. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 6416–6431, 2024.
22. Sheridan Feucht, David Atkinson, Byron C. Wallace, and David Bau. Token Erasure as a Footprint of Implicit Vocabulary Items in LLMs. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 9727–9739, 2024.
23. Somin Wadwha, Adit Krishnan, Runhui Wang, Byron C. Wallace, and Luyang Kong. Learning from Natural Language Explanations for Generalizable Entity Matching. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 6114–6129, 2024.
24. Hye Sun Yun, David Pogrebitskiy, Iain J Marshall, and Byron C Wallace. Automatically extracting numerical results from randomized controlled trials with large language models. In *Proceedings of Machine Learning for Healthcare (MLHC)*, 2024.
25. Jan Trienes, Sebastian Joseph, Jörg Schlötterer, Christin Seifert, Kyle Lo, Wei Xu, Byron C Wallace, and Junyi Jessy Li. Infolossqa: Characterizing and recovering information loss in text simplification. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 4263–4294, 2024.
26. Sebastian Antony Joseph, Lily Chen, Jan Trienes, Hannah Louisa Göke, Monika Coers, Wei Xu, Byron C Wallace, and Junyi Jessy Li. FactPICO: Factuality Evaluation for Plain Language Summarization of Medical Evidence. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 8437–8464, 2024.
27. Hiba Ahsan, Denis Jered McInerney, Jisoo Kim, Christopher A Potter, Geoffrey Young, Silvio Amir, and Byron C Wallace. Retrieving Evidence from EHRs with LLMs: Possibilities and Challenges. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*, 2024.
28. Monica Munnangi, Sergey Feldman, Byron C Wallace, Silvio Amir, Tom Hope, and Aakanksha Naik. On-the-fly Definition Augmentation of LLMs for Biomedical NER. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 3833–3854, 2024.
29. Denis Jered McInerney, William Dickinson, Lucy C. Flynn, Andrea C Young, Geoffrey Young, Jan-Willem van de Meent, and Byron C Wallace. Towards Reducing Diagnostic Errors with Interpretable Risk Prediction. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 7193–7210, 2024.
30. Jiuding Sun, Chantal Shaib, and Byron C. Wallace. Evaluating the Zero-shot Robustness of Instruction-tuned Language Models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. *Spotlight (top 5% of submissions)*.
31. Eric Todd, Millicent Li, Arnab Sen Sharma, Aaron Mueller, Byron C Wallace, and David Bau. LLMs Represent Contextual Tasks as Compact Function Vectors. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
32. Sanjana Ramprasad and Kundan Krishna and Zachary Chase Lipton and Byron C Wallace. Evaluating the Factuality of Zero-shot Summarizers Across Varied Domains. In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 50–59, 2024.
33. Zhaoyue Sun and Gabriele Pergola and Byron C Wallace and Yulan He. Leveraging ChatGPT in Pharmacovigilance Event Extraction: An Empirical Study. In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 344–357, 2024.
34. Sebastian Joseph, Kathryn Kazanas, Keziah Reina, Vishnesh J. Ramanathan, Wei Xu, Byron C. Wallace, and Junyi Jessy Li. Multilingual Simplification of Medical Texts. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 16662–16692, 2023.
35. Koyena Pal, Kiuding Sun, Andrew Yuan, Byron C. Wallace, and David Bau. Anticipating Subsequent Tokens from a Single Hidden State. In *Proceedings of the SIGNLL Conference on Computational Natural Language Learning (CoNLL)*, pages 548–560, 2023.

36. Hye Sun Yun, Iain J. Marshall, Thomas Trikalinos, and Byron C. Wallace. Appraising the Potential Uses and Harms of LLMs for Medical Systematic Reviews. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 10122–10139, 2023.

37. Kundan Krishna, Prakhar Gupta, Sanjana Ramprasad, Byron C. Wallace, Jeffrey P. Bigham, and Zachary C. Lipton. USB: A Unified Summarization Benchmark Across Tasks and Domains. In *Proceedings of Findings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 8826—-8845, 2023.

38. Denis Jered McInerney, Geoffrey Young, Jan-Willem van de Meent, and Byron C. Wallace. CHiLL: Zero-shot Custom Interpretable Feature Extraction from Clinical Notes with Large Language Models. In *Proceeding of Findings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 8477–8494, 2023.

39. Somin Wadhwa, Jay DeYoung, Benjamin Nye, Silvio Amir, and Byron C. Wallace. Jointly extracting interventions, outcomes, and findings from RCT reports with LLMs. In *Proceedings of Machine Learning for Healthcare (MLHC)*, pages 754–771, 2023.

40. Stefán Ólafsson, Paola Pedrelli, Byron C. Wallace, and Timothy Bickmore. Accomodating user expressivity while maintaining safety for a virtual alcohol misuse counselor. In *Proceedings of the Conference on Intelligent Virtual Agents (IVA)*, 2023.

41. Lucy Lu Wang, Yulia Otmakhova, Jay DeYoung, Thinh Hung Truong, Bailey E. Kuehl, Erin Bransom, and Byron C. Wallace. Automated metrics for medical multi-document summarization disagree with human evaluations. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 9871–9889, 2023.

42. Chantal Shaib, Millicent Liaw Li, Sebastian A. Joseph, Iain J. Marshall, Junyi Jessy Li, and Byron C. Wallace. Summarizing, simplifying, and synthesizing medical evidence using gpt-3 (with varying success). In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 1387–1407, 2023.

43. Somin Wadhwa, Silvio Amir, and Byron C. Wallace. Revisiting relation extraction in the era of large language models. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 15566–15589, 2023.

44. Sanjana Ramprasad, Denis Jered McInerney, Iain J. Marshall, and Byron C. Wallace. Automatically summarizing evidence from clinical trials: A prototype highlighting current challenges. In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics (EACL), System Demonstrations*, pages 236–247, 2023.

45. Jinhan Yang, Sarthak Jain, and Byron C. Wallace. How many and which training points would need to be removed to flip this prediction? In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 2571–2584, 2023.

46. Somin Wadhwa, Vivek Khetan, Silvio Amir, and Byron C. Wallace. RedHOT: A corpus of annotated medical questions, experiences, and claims on social media. In *Proceedings of the Findings of the Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 809–827, 2023.

47. Jered McInerney, Geoffrey Young, Jan-Willem van de Meent, and Byron C. Wallace. That's the wrong lung! evaluating and improving the interpretability of unsupervised multimodal encoders for medical data. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 3626–3648, 2022.

48. Zhaoyue Sun, Jiazheng Li, Gabriele Pergola, Byron C. Wallace, Bino John, Nigel Greene, Joseph Kim, and Yulan He. Phee: A dataset for pharmacovigilance event extraction from text. In *Proceedings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, 2022.

49. Sarthak Jain, Varun Manjunatha, Byron C. Wallace, and Ani Nenkova. Influence functions for sequence tagging models. In *Proceedings of the Findings of the Conference on Empirical Methods for Natural Language Processing (EMNLP)*, pages 824–839, 2022.

50. Nikita Salkar, Thomas Trikalinos, Byron C. Wallace, and Ani Nenkova. Self-repetition in abstractive neural summarizers. In *Proceedings of the Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)*, pages 341–350, 2022.

51. Pouya Pezeshkpour, Sarthak Jain, Sameer Singh, and Byron C. Wallace. Combining Feature and Instance Attribution to Detect Artifacts. In *Proceedings of the Findings of the Conference of the Association for Computational Linguistics (ACL)*, pages 1934–1946, 2022.

52. Ashwin Devaraj, William Berkeley Sheffield, Byron C Wallace, and Junyi Jessy Li. Evaluating Factuality in Text Simplification. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 7331–7345, 2022. *ACL 2022 Outstanding Paper*.

53. David Lowell, Brian Howard, Zachary C. Lipton, and Byron C. Wallace. Unsupervised Data Augmentation with Naive Augmentation and without Unlabeled Data. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

54. Xiongyi Zhang, Jan-Willem van de Meent, and Byron Wallace. Disentangling representations of text by masking transformers. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 778–791, 2021.

55. Diego Garcia-Olano, Yasumasa Onoe, Ioana Baldini, Joydeep Ghosh, Byron Wallace, and Kush Varshney. Biomedical interpretable entity representations. In *Proceedings of the Findings of the Association for Computational Linguistics (ACL)*, pages 3547–3561, Online, 2021. Association for Computational Linguistics.

56. Pouya Pezeshkpour, Sarthak Jain, Byron Wallace, and Sameer Singh. An empirical comparison of instance attribution methods for NLP. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 967–975, Online, June 2021. Association for Computational Linguistics.

57. Silvio Amir, Jan-Willem van de Meent, and Byron C. Wallace. On the Impact of Random Seeds on the Fairness of Clinical Classifiers. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 3808–3823, 2021.

58. Ashwin Devaraj, Iain Marshall, Byron C. Wallace, and Junyi Jessy Li. Paragraph-level simplification of medical texts. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4972–4984, Online, June 2021. Association for Computational Linguistics.

59. Eric Lehman, Sarthak Jain, Karl Pichotta, Yoav Goldberg, and Byron C. Wallace. Does BERT Pretrained on Clinical Notes Reveal Sensitive Data? In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 946–959, 2021.

60. Benjamin E. Nye, Jay DeYoung, Eric Lehman, Ani Nenkova, Iain J. Marshall, and Byron C. Wallace. Understanding Clinical Trial Reports: Extracting Medical Entities and Their Relations. In *Proceedings of AMIA Informatics Summit*, 2021. *Recipient of the Best Student Paper Award*.

61. Byron C. Wallace, Sayantan Saha, Frank Soboczenski, and Iain J. Marshall. Generating (Factual?) Narrative Summaries of RCTs: Experiments with Neural Multi-Document Summarization. In *Proceedings of AMIA Informatics Summit*, 2021.

62. Denis Jered McInerney, Borna Dabiri, Anne-Sophie Touret, Geoffrey Young, Jan-Willem van de Meent, and Byron C. Wallace. Query-Focused EHR Summarization to Aid Imaging Diagnosis. In *Proceedings of Machine Learning in Health Care (MLHC)*, 2020.

63. Somin Wadhwa, Kanhua Yin, Kevin S. Hughes, and Byron C. Wallace. Semi-Automating Knowledge Base Construction for Cancer Genetics. In *Proceedings of the Conference on Automated Knowledge Base Construction (AKBC)*, 2020.

64. Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. ERASER: A Benchmark to Evaluate Rationalized NLP Models. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 4443–4458, 2020.

65. Sarthak Jain, Sarah Wiegreffe, Yuval Pinter, and Byron C. Wallace. Learning to Faithfully Rationalize by Construction. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 4459–4473, 2020.

66. Benjamin Nye, Ani Nenkova, Iain J. Marshall, and Byron C. Wallace. Trialstreamer: Mapping and Browsing Medical Evidence in Real-Time. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL), System Demonstrations*, pages 63–69, 2020.

67. Xiaochuang Han, Byron C. Wallace, and Yulia Tsvetkov. Explaining Black Box Predictions and Unveiling Data Artifacts through Influence Functions. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 5553–5563, 2020.

68. Eric Lee, Byron C. Wallace, Karla Galaviz, and Joyce C. Ho. MMiDaS-AE: Multi-modal Missing Data aware Stacked Autoencoder for Biomedical Abstract Screening. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*, pages 139–150, 2020.

69. Stefan Olafsson, Byron C. Wallace, and Timothy Bickmore. Towards a Computational Framework for Automating Substance Use Counseling with Virtual Agents. In *Proceedings of the International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, pages 966–974, 2020.

70. David Lowell, Zachary Lipton, and Byron C. Wallace. Practical Obstacles to Deploying Active Learning. In *Proceedings of Empirical Methods in Natural Language Processing (EMNLP)*, pages 21–30, 2019.

71. Avijit Thawani, Michael J. Paul, Urmimala Sarkar, and Byron C. Wallace. Are Online Reviews of Physicians BiasedAgainst Female Providers? In *Proceedings of Machine Learning in Health Care (MLHC)*, pages 406–423. Journal of Machine Learning Research (JMLR) Conferences Track, 2019.

72. Sarthak Jain and Byron C. Wallace. Attention is not Explanation. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 3543–3556, 2019.

73. Yinfei Yang, Oshin Agarwal, Chris Tar, Byron C. Wallace, and Ani Nenkova. Predicting Annotation Difficulty to Improve Task Routing and Model Performance for Biomedical Information Extraction. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 1471–1480, 2019.

74. Eric Lehman, Jay DeYoung, Regina Barzilay, and Byron C. Wallace. Inferring Which Medical Treatments Work from Reports of Clinical Trials. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 3705–3717, 2019.

75. Babak Esmaeili, Hongyi Huang, Byron Wallace, and Jan-Willem van de Meent. Structured neural topic models for reviews. In *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATs)*, pages 3429–3439, 2019.

76. An Thanh Nguyen, Matthew Lease, and Byron C. Wallace. Explainable Modeling of Annotations in Crowdsourcing. In *Proceedings of the Annual ACM Intelligent User Interfaces (IUI) conference*, pages 575–579, 2019.

77. Ramin Mohammadi, Sarthak Jain, Stephen Agboola, Ramya Palacholla, Sagar Kamarthi, and Byron C. Wallace. Learning to Identify Patients at Risk of Uncontrolled Hypertension Using Electronic Health Records Data. In *Proceedings of the AMIA Informatics Summit*, 2019.

78. Sarthak Jain, Edward Banner, Jan-Willem van de Meent, Iain J. Marshall, and Byron C. Wallace. Learning Disentangled Representations of Texts with Application to Biomedical Abstracts. In *Proceedings of Empirical Methods in Natural Language Processing (EMNLP)*, page 4683–4693, 2018.

79. Gaurav Singh, James Thomas, Iain J. Marshall, John Shawe-Taylor, and Byron C. Wallace. Structured Multi-Label Biomedical Text Tagging via Attentive Neural Tree Decoding. In *Proceedings of Empirical Methods in Natural Language Processing (EMNLP)*, pages 2837–2842, 2018.

80. An Thanh Nguyen, Byron C. Wallace, and Matthew Lease. Believe it or not: Designing a Human-AI Partnership for Mixed-Initiative Fact-Checking. In *Proceedings of the ACM User Interface Software and Technology Symposium (UIST)*, pages 189–199, 2018.

81. Benjamin Nye, Jessy Li, Roma Patel, Yinfei Yang, Iain Marshall, Ani Nenkova, and Byron C. Wallace. A Corpus with Multi-Level Annotations of Patients, Interventions and Outcomes to Support Language Processing for Medical Literature. In *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*, pages 197–207, 2018.

82. Roma Patel, Yinfei Yang, Iain Marshall, Ani Nenkova, and Byron C. Wallace. Syntactic Patterns Improve Information Extraction for Medical Search. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 371–377, 2018.

83. An Thanh Nguyen, Aditya Kharosekar, Matthew Lease, and Byron C. Wallace. An Interpretable Joint Graphical Model for Fact-Checking from Crowds. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, pages 1511–1518, 2018.

84. Ye Zhang and Byron C. Wallace. A Sensitivity Analysis of (and Practitioners' Guide to) Convolutional Neural Networks for Sentence Classification. *International Joint Conference on Natural Language Processing (IJCNLP)*, pages 253–263, 2017.

85. Gaurav Singh, Iain Marshall, James Thomas, John Shawe-Taylor, and Byron C. Wallace. A Neural Candidate-Selector Architecture for Automatic Structured Clinical Text Annotation. *International Conference on Information and Knowledge Management (CIKM)*, pages 1519–1528, 2017.

86. Silvio Amir, Glen Coppersmith, Paula Carvalho, Mario J. Silva, and Byron C. Wallace. Quantifying Mental Health from Social Media with Neural User Embeddings. In *Proceedings of Machine Learning in Health Care (MLHC)*, pages 306–321. Journal of Machine Learning Research (JMLR) Conferences Track, 2017.

87. Iain Marshall, Joël Kuiper, Edward Banner, and Byron C. Wallace. Automating Biomedical Evidence Synthesis: RobotReviewer. In *Proceedings of the Association for Computational Linguistics (ACL), System Demonstrations*, pages 7–12. Association for Computational Linguistics (ACL), 2017.

88. Ye Zhang, Matthew Lease, and Byron C. Wallace. Exploiting Domain Knowledge via Grouped Weight Sharing with Application to Text Categorization. In *Proceedings of the Association for Computational Linguistics (ACL)*, pages 155–160. Association for Computational Linguistics (ACL), 2017.

89. An Thanh Nguyen, Byron C. Wallace, Junyi Jessy Li, Ani Nenkova, and Matthew Lease. Aggregating and Predicting Sequence Labels from Crowd Annotations. In *Proceedings of the Association for Computational Linguistics (ACL)*, pages 299–309. Association for Computational Linguistics (ACL), 2017.

90. Zhiguo Yu, Byron C. Wallace, Todd Johnson, and Trevor Cohen. Retrofitting Concept Vector Representations of Medical Concepts to Improve Estimates of Semantic Similarity and Relatedness. In *Proceedings of MEDINFO*, pages 657 – 661. International Medical Informatics Association (IMIA), 2017.

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Workshop/symposium publications & technical reports

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157. Alberto Mario Ceballos-Arroyo, Monica Munnangi, Jiuding Sun, Karen Zhang, Jered McInerney, Byron C. Wallace, and Silvio Amir. Open (clinical) llms are sensitive to instruction phrasings. In *Proceedings of the BioNLP Workshop*, Bangkok, Thailand, 2024. Association for Computational Linguistics.

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161. Gregory Kell, Iain Marshall, Byron C. Wallace, and Andre Jaun. What Would it Take to get Biomedical QA Systems into Practice? In *Proceedings the Workshop on Machine Reading for Question Answering (MRQA) at EMNLP*, 2021.

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165. An Thanh Nguyen, Matthew Lease, and Byron C. Wallace. MASH: software tools for developing interactive and transparent machine learning systems. In *Proceedings of ACM IUI Workshop on Explainable Smart Systems (ExSS)*, 2019.

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167. Gaurav Singh, Iain Marshall, James Thomas, and Byron C. Wallace. Identifying Diagnostic Test Accuracy Publications using a Deep Model. In *CLEF eHealth*, 2017.

168. Ryan Bridges, Jette Henderson, Joyce Ho, Byron C Wallace, and Joydeep Ghosh. Automated Verification of Phenotypes using PubMed. In *Proceedings of the Workshop on Methods and Applications in Healthcare Analytics at ACM-BCB*, 2016.

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174. Byron C. Wallace, Michael J. Paul, and Noemie Elhadad. What Predicts Media Coverage of Health Science Articles? In *Proceedings of the International Workshop on the World Wide Web and Public Health Intelligence (W3PHI)*. Association for the Advancement of Artificial Intelligence (AAAI), 2015.

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Book Chapters

177. Timothy Bickmore and Byron C. Wallace. Intelligent agents and dialog systems. In Trevor A Cohen, Vimla L Patel, and Edward H Shortliffe, editors, *Intelligent Systems in Medicine and Health: The Role of AI*, chapter 9. Springer, 2022.

178. Byron C. Wallace, Issa J. Dahabreh, Christopher H. Schmid, Joseph Lau, and Thomas A. Trikalinos. Modernizing evidence synthesis for evidence-based medicine. In Robert A. Greenes, editor, *Clinical Decision Support (Second Edition)*, chapter 12. Wiley, 2013.

Commentaries & editorials

179. Byron C. Wallace. *Invited Early Career Spotlight Extended Abstract*: What Does the Evidence Say? Models to Help Make Sense of the Biomedical Literature. *International Joint Conference on Artificial Intelligence*, pages 6416–6420, 2019.

180. Jenna Wiens and Byron C. Wallace. Editorial: Special Issue on Machine Learning for Health and Medicine. *Machine Learning Journal*, pages 1–3, 2015.

181. Byron C. Wallace and Iain J. Marshall. Invited Response: Using Text Mining for Study identification in Systematic Reviews: a Systematic Review of Current Approaches. *Cochrane Methods*, 2015.

182. Julian Elliott, Ida Sim, Jessica Thomas, Nancy Owens, Gordon Dooley, Jacob Riis, Byron Wallace, James Thomas, Anna Noel-Storr, Gabriel Rada, Caroline Struthers, Tracey Howe, Harriet MacLehose, Linn Brandt, Ilkka Kunnamo, and Chris Mavergames. #CochraneTech: Technology and the Future of Systematic Reviews. *The Cochrane Library*, 2014.

183. Byron C. Wallace and Thomas A. Trikalinos. Invited Response: Applications of text mining within systematic reviews. *Cochrane Methods*, 2012.

Selected invited talks & panels

Note that this is non-exhaustive and that these are in addition to the talks accompanying the conference publications above.

1. What (if anything) can mechanistic interpretability do for healthcare? *McGill AI & Healthcare Reading Group*, 10/13, 2026.
2. What (if anything) can mechanistic interpretability do for healthcare? *Keynote at the UMich AI and Health Symposium*, 9/11, 2026.
3. What (if anything) can mechanistic interpretability do for healthcare? *Northeastern Institute for Experential AI Faculty Seminar*, 5/12, 2026.
4. Evaluation Frameworks for AI. *Finding What Works in Health Care: Updating Standards for Systematic Reviews - Webinar Series*, 5/1, 2026.
5. What (if anything) can mechanistic interpretability do for healthcare? *Tufts Computer Science Colloquia*, 4/2, 2026.
6. Some potential uses for and (current) limitations of mechanistic interpretability methods. *UPenn, Computational Linguistics lunch (CLunch)*, 9/23, 2025.
7. What (if Anything) can Interpretability do for Healthcare? *Keynote at the ICML Workshop on Actionable Interpretability*, 7/19, 2025.
8. Large Language Models: A Quick Primer. *Practical Uses of AI in Home Hospital Patient Identification Workshop at Northeatern*, 6/9, 2025.
9. LLMs for healthcare: Risks and interpretability methods to (possibly) mitigate them. *Dartmouth Computer Science Colloquium*, 5/9, 2025.
10. LLMs for healthcare: Risks and interpretability methods to (possibly) mitigate them. *UMass Lowell Computer Science Colloquium*, 3/21, 2025.
11. 'The title may be dull, but it is a very good description', or, some of eugene's invaluable advice on interdisciplinary nlp research. *Eugene Charniak Memorial, Brown University*, 11/8, 2024.
12. *Invited panel member*: Underloved innovations in Health AI (other panelists: Hamsa Bastani, Matthew McDermott, Irene Trampoline; Moderator: Marzyeh Ghassemi). *ACM Conference on Health, Inference, and Learning (CHIL)*, 6/24, 2024.
13. Helping physicians make sense of medical evidence with Large Language Models. *MGH Genetics Lecture*, 6/5, 2024.
14. Potential uses (and risks) of LLMs for summarizing evidence and aiding clinical decision making. *Plenary "Ted-Style" Talk at the Annual Clinical Genetics Meeting (ACGM)*, 3/14, 2024.
15. Helping physicians make sense of medical evidence with Large Language Models. *AI/ML Community of Practice Speaker Series at Takeda*, 1/23, 2024.
16. Helping physicians make sense of medical evidence with Large Language Models. *Precision Psychiatry Seminar, Massachusetts General Hospital*, 12/15, 2023.
17. Interrogating zero- and few-shot learners. *University of Chicago / TTIC NLP Seminar*, 12/8, 2023.

18. Helping physicians make sense of medical evidence with Large Language Models. *ML+Health Seminar Series at MIT*, 9/15, 2023.
19. *Panel moderator* Machine Learning for Healthcare in the Era of ChatGPT. *ACM Conference on Health, Inference, and Learning (CHIL)*, 6/24, 2023.
20. Large language models for extracting medical evidence. *King's College London Computer Science Seminar*, 6/15, 2023.
21. Large language models for extracting medical evidence. *King's College London Public Health Seminar*, 3/29, 2023.
22. A few fragilities of neural summarizers: Repetition and synthesis. *Robustness in Sequence Modeling Workshop at NeurIPs*, 12/2, 2022.
23. Learning to synthesize medical evidence: Challenges in conditional text generation for healthcare. *Industrial Transformation Training Centre in Cognitive Computing for Medical Technologies Seminar Series at the University of Melbourne*, 4/26, 2022.
24. Learning to synthesize medical evidence: Challenges in conditional text generation for healthcare. *Distinguished Researcher Presentation at Accenture Labs*, 4/26, 2022.
25. Methods to aid model debugging: From rationales to influence. *Responsible Data Science and AI Speaker Series at UIUC*, 4/22, 2022.
26. Learning to synthesize medical evidence: Challenges in conditional text generation for healthcare. *AI Seminar at gRED, Genentech*, 4/14, 2022.
27. Challenges in conditional text generation for health: Learning to summarize and simplify medical evidence. *UTHSC-ORNL Center for Biomedical Informatics Lecture Series*, 3/8, 2022.
28. Challenges in conditional text generation for health: Learning to summarize and simplify medical evidence. *Boston Children's Hospital Artificial Intelligence and Machine Learning Working Group: Journal Club*, 12/14, 2021.
29. *Invited panel member*: Applying state of the art language models to enable better clinical natural language processing (other panelists: Emily Alsentzer, Adam Wright, Bryan Steitz, Hoo Chang Shin). *AMIA Annual Symposium*, 11/3, 2021.
30. Artificial intelligence for systematic reviews. *Machine Learning in Health: Opportunities and Challenges Speaker Series at McMaster University*, 10/28, 2021.
31. *Invited panel member*: Learning in healthcare. *INFORMS Annual Meeting*, 10/25, 2021.
32. Machine learning aided risk of bias assessment. *Cochrane US Virtual Methods Symposium*, 2/23, 2021.
33. What does the evidence say? Models to help make sense of the biomedical literature. *Keynote at the Scientific Document Understanding Workshop at AAAI*, 2/9, 2021.
34. What does the evidence say? Models to help make sense of the biomedical literature. *NYU NLP/Text-as-Data Speaker Series*, 2/4, 2021.
35. Learning to Summarize Medical Evidence. *Sheffield NLP Seminar Series* (Remote due to COVID-19), 11/5, 2020.
36. Learning to Summarize Medical Evidence. *Memorial Sloan Kettering: Computational Oncology Seminar Series* (Remote due to COVID-19), 10/20, 2020.
37. Learning to Summarize Medical Evidence. *Pitt-CMU Seminar MLxMED: Machine Learning in Medicine Seminar* (Remote due to COVID-19), 10/14, 2020.
38. Learning to Summarize Medical Evidence. *Computational Linguistics and Information Processing at Maryland (CLIP) seminar, University of Marlyand (UMD)* (Remote due to COVID-19), 9/16, 2020.
39. Using rationales and influential training examples to (attempt to) explain neural predictions in NLP. *Machine Learning at Georgia Tech Seminar* (Remote due to COVID-19), 9/9, 2020.
40. What does the evidence say? Models to help make sense of the biomedical literature. *SciNLP: Natural Language Processing and Data Mining for Scientific Text. Co-located with AKBC 2020: https://scinlp.org/.* (Remote due to COVID-19), 6/25, 2020.
41. Towards interpretable neural models for NLP. *Workshop on Mind and Machine Intelligence at UCSB*, 2/21, 2020.
42. A few case studies on NLP applied to social media for health: Evaluating bias and establishing trust . *Promoting Research in Social Media and Health Conference (PRISM)*, 12/6, 2019.
43. What does the evidence say? Models to help make sense of the biomedical literature. *UMass Amherst, Machine Learning and Friends Lunch (MLFL) series*, 9/26, 2019.
44. What does the evidence say? Models to help make sense of the biomedical literature. *UMass Lowell Computer Science Colloquium*, 9/18, 2019.

45. What does the evidence say? Models to help make sense of the biomedical literature. *Carnegie Mellon Language Technologies Institute (LTI) Colloquium*, 8/30, 2019.
46. What does the evidence say? Models to help make sense of the biomedical literature. *IJCAI Early Career Spotlight*, 8/15, 2019.
47. What does the evidence say? Models to help make sense of the biomedical literature. *Salesforce Research Speaker Series*, 7/19, 2019.
48. Extracting structured evidence from reports of clinical trials. *Bloomberg Speaker Series*, 6/21, 2019.
49. Machine learning and natural language processing for health: From automating evidence synthesis to making clinical predictions. *Johns Hopkins University, Graduate Summer Institute Seminar Series in Epidemiology and Biostatistics*, 6/13, 2019.
50. On the Use and Interpretation of Neural Attention Mechanisms in NLP. *Advances in Data Sciences*, 5/20, 2019.
51. *Invited panel member*: Machine learning and human language (other panelists: Steve Bethard, Robert Kraut; chair: Dragomir radev). *American Association for the Advancement of Science (AAAS) Annual Meeting*, 2/16, 2019.
52. ML and NLP for unstructured health data. *Wolters Kluwer*, 11/8, 2018.
53. Semi-Automating Biomedical Evidence Synthesis via ML and NLP. *Computational Health Informatics Program (CHIP) Seminar at Boston Children's Hospital*, 12/13, 2018.
54. Training Neural NLP Models in Minimally Supervised Settings. *Applied and Interdisciplinary Mathematics Seminar, Northeastern University*, 10/9, 2018.
55. Keynote: Making Sense of the Biomedical Literature via ML and NLP. *The Joint Workshop on Bibliometric-enhanced IR and NLP for Digital Libraries at SIGIR*, 7/12, 2018.
56. Training Neural NLP Models in Minimally Supervised Settings. *New England Machine Learning Day (NEML)*, 5/7, 2018.
57. NLP in literature mining: Semi-automated quality assessment. *IBTN Workshop on Intelligent Interventions*, 5/26, 2018.
58. Making Sense of the Biomedical Literature via Machine Learning and Natural Language Processing. *Open Data Science Conference*, 5/4, 2018.
59. *Invited panel member*: Explainable AI models for health informatics (other panelists: David Sontag, Ziad Obermeyer, Anil Jain). *Northeast Computational Health Summit (NECHS)*, 4/27, 2018.
60. Expediting Clinical Evidence Synthesis via Machine Learning, Natural Language Processing and Crowdsourcing. *Keynote at the 4th International Symposium on Systematic Review and Meta-Analysis of Laboratory Animal Studies*, 08/24, 2017.
61. Expediting Clinical Evidence Synthesis via Machine Learning, Natural Language Processing and Crowdsourcing. *Weill Cornell Medicine, Machine Learning in Medicine series*, 07/14, 2017.
62. Expediting Clinical Evidence Synthesis via Machine Learning, Natural Language Processing and Crowdsourcing. *MIT (MEDG Reading Group)*, 06/02, 2017.
63. *Invited panel member*: Data mining, clinical medicine, and medical research: The current state and future of the nexus (other panelists: Genevera Allen, Elmer V Bernstam, Chris Jermaine, Olivier Lichtarge, Srinivasan Parthasarathy, Jimeng Sun). *Statistics and Data Mining (SDM)*, 4/28, 2017.
64. Expediting Clinical Evidence Synthesis via Machine Learning, Natural Language Processing and Crowdsourcing. *UMass, Worcester Medical School (BioNLP group)*, 01/05, 2017.
65. Data Science, Social Media and Health. *Promoting Research in Social Media and Health Symposium (PRISM Health Symposium), UCSF, San Francisco, CA*, 12/02, 2016.
66. Some recent work on Convolutional Neural Networks (CNNs) for text classification. *Raytheon BBN Technologies (BBN), Cambridge, MA*, 11/22, 2016.
67. Expediting Clinical Evidence Synthesis via Machine Learning, Natural Language Processing and Crowdsourcing. *Amazon Machine Learning Lecture Series, Seattle, WA*, 11/17, 2016.
68. Expediting Clinical Evidence Synthesis via Machine Learning, Natural Language Processing and Crowdsourcing. *Evidence Synthesis: Current Practices and Future Possibilities (Panel at IEEE ICHI), Chicago, IL*, 10/5, 2016.
69. Expediting Clinical Evidence Synthesis via Machine Learning, Natural Language Processing and Crowdsourcing. *Penn Institute for Computational Science (PICS) Symposium on Emerging Paradigms in Scientific Discovery, Philadelphia, PA*, 10/6, 2016.
70. Open Challenges in Automating Clinical Evidence Synthesis. *International Conference on the Automation of Systematic Reviews, Philadelphia, PA*, 10/3, 2016.

71. Some recent work on Convolutional Neural Networks (CNNs) for text classification. *MIT Computer Science and Artificial Intelligence Laboratory, Cambridge, MA*, 10/11, 2016.
72. Some recent work on Convolutional Neural Networks (CNNs) for text classification. *University of Lisbon, Portugal*, 6/22, 2016.
73. Automating Evidence Synthesis via Machine Learning and Natural Language Processing. *Machine Learning and Data Analytics Symposium (MLDAS), Doha, Qatar*, 3/15, 2016.
74. Automating evidence synthesis via machine learning and natural language processing. *Symposium on Text Mining, Bristol University*, 11/11, 2015.
75. Automating evidence synthesis via machine learning and natural language processing. *Department of Biostatistics, Brown University*, 10/19, 2015.
76. Automating evidence synthesis via machine learning and natural language processing. *Department of Computer and Information Science, University of Pennsylvania*, 3/31, 2015.
77. Automating evidence synthesis. *Department of Biomedical Informatics, University of Texas at Houston*, 10/1, 2014.
78. Can machine learning help us improve physician communication? *Department of Biomedical Informatics, Columbia University, NYC*, 5/15, 2014.
79. Automating the systematic review: Tools and methods. *National Institute of Environmental Health Sciences, Durham, North Carolina*, 4/30, 2014.
80. Machine Learning in Evidence-Based Medicine: Managing the Clinical Data Deluge. *University of Groningen, Netherlands*, 4/24, 2014.
81. Can Machine Learning Help us Improve Physician Communication? *MIT Lincoln Lab, Lexington, MA*, 11/1, 2013.
82. Can Machine Learning Help us Improve Physician Communication? *Data Management Lab at Boston University, Boston, MA*, 11/8, 2013.
83. Semi-Automating Systematic Reviews: Text mining, NLP and machine learning. *#CochraneTech Symposium: Technology and the future of the systematic review, Quebec City, Canada*, 9/19, 2013.
84. Statistical Models of Patient-Doctor Communication. *Meaningful Use of Complex Medical Data (MUCMD), LA, CA*, 8/16, 2013.
85. Machine learning in health informatics: Making better use of domain experts (doctoral dissertation awards session presentation). *The 19th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Chicago, IL*, 8/12, 2013.
86. Machine learning in systematic reviews: Making better use of domain expertise. *Statistical Analysis of "Big Data" Group, Brown University, Providence, RI*, 11/09, 2012.
87. Better models, less effort: Active learning and dual supervision. *Department of Computer Science at UMass, Boston*, 03/07, 2012.

Research Support (Grants)

A few notes on the below

- *Listed in reverse chronological order, w.r.t start date.*
- *I **bold** the titles of grants on which I am Principal Investigator (PI) or Co-/M- PI.*
- *Dollar amounts are best estimates of total direct award figures. For active awards, these are projections. In cases where grants involve multiple PIs (Co-PIs or M-PIs), I note my estimated portion parenthetically.*

Grant Title **Mechanistic Interpretability for Healthcare**
Funder Open Philanthropy
Role Principal Investigator
Period 9/1/2025–8/31/2027 (*grant agreement finalized*)
Amount 800k (Wallace)

Grant Title **SOLACE-AI: Synthesis of Online Literature for Adaptation to Climate-Change Emergencies'**

Funder Wellcome Trust
Role Principal Investigator
Period 3/1/2025–2/28/2030
Amount ∼5m (Total) / ∼810k (Wallace)
Collaborators PI/PD Iain Marshall (King's College), PI Thomas Trikalinos (Brown University), PI Nasir Warfa (Jigjiga University), PI Shreya Shrikhande (World Heart Federation)

Grant Title **Building Safety Guards into LLMs for Trustworthy Automatic Simplification of Medical Documents (R01LM014600)**
Funder National Institutes of Health (NIH) / National Library of Medicine (NLM)
Role Principal Investigator
Period 9/18/2024–8/31/2028
Amount ∼1m (Total) / ∼320k (Wallace)
Collaborators PI/PD Jessy Li (UT Austin), PI Wei Xu (Georgia Tech)

Grant Title **National Deep Inference Fabric for Very Large Language Models (NSF 2408455)**
Funder National Science Foundation (NSF)
Role Co-Principal Investigator
Period 5/1/2024–4/30/2028
Amount ∼$8.99m (Total)
Collaborators PI/PD David Bau, Co-PI Arjun Guha, Co-PI Jonathan Bell, Co-PI Carla Brodley

Grant Title **Targeted Neural Text Summarization of Electronic Medical Records to Improve Imaging Diagnostics (R01LM013772)**
Funder National Institutes of Health (NIH) / National Library of Medicine (NLM)
Role Project Director / Principal Investigator (Contact) (PD/PI)
Period 9/02/2022–8/31/2025
Amount ∼$630k (total); ∼$330k (Wallace)
Collaborators MPI Geoffrey Young (BWH)

Grant Title **Expert-in-the-Loop Neural Summarization for Consequential Domains (NSF 2211954)**
Funder National Science Foundation (NSF)
Role Project Director (PD) / Principal Investigator (PI)
Program CISE/RI Medium
Period 7/2022 – 6/2026
Amount ∼$1.2m (total); ∼$600k (Wallace)
Collaborators Co-Principal Investigator Zachary Lipton (CMU)

Grant Title **General research support** (multiple gifts)
Funder Adobe Research (`https://research.adobe.com/`)
Role Principal Investigator (PI)
Period 9/2021, 6/22, 9/2023 (unrestricted gift)
Amount 45k (total)

Grant Title **Identifying Medical Misinformation on Social Media**
Funder Accenture Research

Role Principal Investigator (PI)
Period 6/2021 (unrestricted gift)
Amount 75k

Grant Title Safety Promotion through Early Event Detection in the Elderly (SPEEDe; R01AG062499)
Funder National Institutes of Health
Role Investigator
Collaborators PI Adam Wright (Vanderbilt)
Period 3/1/2020 – 3/1/2024
Amount ∼$80k (Wallace)

Grant Title **Semi-Automating Data Extraction for Systematic Reviews (Renewal; 2R01LM012086)**
Funder National Institutes of Health/National Library of Medicine
Role Project Director/Principal Investigator (Contact) (PD/PI)
Collaborators MPI Iain J. Marshall (King's College), Co-I Thomas Trikalinos (Brown)
Period 9/1/2019 – 8/31/2023
Amount ∼$1.2M (total); ∼$520k (Wallace)

Grant Title **Learning Disentangled Representations for Text to Aid Interpretability and Transfer (NSF 1901117)**
Funder National Science Foundation (NSF)
Role Principal Investigator (PI)
Program CISE/RI Medium
Period 9/2019 – 9/2022
Amount ∼$1m (total); ∼$500k (Wallace)
Collaborators Co-Principal Investigator Jan-Willem van de Meent (Northeastern)

Grant Title Research and Development of an Open, Extensible, Web-Based Information Extraction Workbench for Systematic Review (R43ES029901)
Funder National Institutes of Health (NIH)
PI Ruchir Shah (Sciome LLC)
Role Investigator
Program Small Business Innovation Research (SBIR) Grant
Period 1/2019 – 7/2019
Amount ∼$150k (total); ∼$55k (Wallace)

Grant Title **CAREER: Structured Scientific Evidence Extraction: Models and Corpora (NSF 1750978)**
Funder National Science Foundation (NSF)
Role Principal Investigator (PI)
Program CISE/III
Period 7/2018 – 6/2023
Amount ∼550k

Grant Title **Neural Models for Text: Improving Efficiency, Interpretability and Accuracy (ARO W911NF1810328)**

Funder	Army Research Office (ARO)
Role	Principal Investigator (PI)
Period	7/2018 – 7/2021
Amount	∼415k (total); ∼366k (Wallace)
Collaborators	Investigator Matt Lease (UT Austin)

Grant Title	Measuring, Mining, and Understanding Communication Behaviors: Markers for Quality Healthcare (VA HX002289)
Funder	VA Health Services Research and Development Service
PI	Thomas Houston (UMass Medical School)
Role	Investigator
Period	9/2018 – 1/2021
Amount	∼$1.1m (total); ∼$32k (Wallace)

Grant Title	**Neural Embeddings for Inferring Mental Health Status from Twitter Streams**
Funder	Qntfy (`https://qntfy.com/`)
Role	Principal Investigator (PI)
Period	5/2017 (unrestricted gift)
Amount	∼$20k

Grant Title	**Hybrid Approaches to Optimizing Evidence Synthesis via Machine Learning and Crowdsourcing (R03HS025024)**
Funder	Agency for Healthcare Research and Quality (AHRQ)
Role	Principal Investigator (PI)
Program	Small Research Grants; Special Emphasis on Innovative Methods Research to Increase the Utility of Systematic Reviews
Period	9/30/2016 – 9/29/2017
Amount	∼$100k

Grant Title	**Crowdsourcing Mark-up of the Medical Literature to Support Evidence-Based Medicine and Develop Automated Annotation Capabilities (UH2CA203711)**
Funder	National Institutes of Health/National Cancer Institute
Role	Project Director/Principal Investigator (Contact) (PD/PI)
Collaborators	MPIs Ani Nenkova and Zachary Ives (UPenn)
Program	Big Data to Knowledge (BD2K)
Period	8/1/2016 – 9/29/2018
Amount	∼$450k (total); ∼$210k (Wallace)

Grant Title	**Semi-Automating Data Extraction for Systematic Reviews (R01LM012086)**
Funder	National Institutes of Health/National Library of Medicine
Role	Project Director/Principal Investigator (Contact) (PD/PI)
Collaborators	MPIs Iain J. Marshall (King's College), Thomas Trikalinos (Brown), Randolph Bias (UT Austin)
Period	9/20/2015 – 8/31/2019
Amount	∼$1.18M (total); ∼$625k (Wallace)

Grant Title	**Manual and Automatic Analysis of Patients' Values and Preferences Using Seton HCAHPS Surveys**
Funder	Seton Medical Center/UT Center for Health and Social Policy (CHASP)
Role	Co-Principal Investigator (Co-PI)
Collaborators	Co-PIs Kenneth Fleischmann and Bo Xie (UT)
Period	8/2015 – 9/2016
Amount	∼$50k (total); ∼$10k (Wallace)

Grant Title	Expression and Recognition of Irony in Multicultural Social Media (UTAP-EXPL/EEI-ESS/0031/2014)
Funder	FCT
Role	Investigator
PI	Paula Carvalho (INESC-ID Lisboa & Universidade Europeia)
Collaborators	Investigators Mário J. Silva and Bruno Martins (University of Lisbon)
Period	1/2015 – 1/2016
Amount	∼$35k (Wallace)

Grant Title	Influencing Cervical Cancer Prevention and Detection Online through Social Media (R01CA178875)
Funder	National Institutes of Health (NIH)/National Cancer Institute (NCI)
Role	Investigator
PI	MPIs Urmimala Sarkar (UCSF) and Damon Centola (UPenn)
Period	9/2014 – 9/2017
Amount	∼$1.3m (total); ∼$92k (Wallace)

Grant Title	**Scaling Evidence-Based Medicine via Automation and Crowdsourcing**
Funder	Brown University
Role	Principal Investigator (PI)
Collaborators	Tim Kraska, Ugur Cetintemel and Thomas Trikalinos (Brown)
Program	Seed Funding (Internal to Brown)
Period	8/2014 – 8/2015
Amount	∼$80k

Grant Title	**Sociolinguistically Informed Natural Language Processing: Automating Irony Detection (W911NF1410442)**
Funder	Army Research Office (ARO)
Role	Principal Investigator (PI)
Collaborators	Investigators Eugene Charniak, Thomas Trikalinos and Laura Kertz (Brown)
Program	Mathematical Sciences Division
Period	9/2013 – 9/2016
Amount	∼$300k

Grant Title	**Making Advanced Statistical Tools Accessible for Quantitative Research Synthesis and Discovery in Ecology and Evolutionary Biology (DBI-1262442)**
Funder	National Science Foundation (NSF)
Role	Co-Principal Investigator (Co-PI)

Collaborators	Co-PI Jessica Gurevitch (SUNY), Co-PI Marc Lajeunesse (USF), Co-PIs Thomas Trikalinos and Christopher H. Schmid (Brown)
Program	Collaborative research: ABI Development: Making Advanced Statistical Tools Accessible for Quantitative Research Synthesis and Discovery in Ecology and Evolutionary Biology.
Period	5/1/2013 – 5/1/2016
Amount	∼$915k (total); ∼$500k (Wallace)

Other support I have twice received support from the Amazon AWS Cloud Credits for Research program. Once in 2014 (for $2k) and again in 2016 (for $4.5k)

Teaching

Northeastern

Spring 2026 CS7180, Actionable Interpretability (Graduate Seminar)
Enrollment: 26
Mean overall instructor teaching effectiveness rating: 4.7/5.0

Fall 2024 DS4440, Practical Neural Networks (Advanced Undergraduate)
Enrollment: 8
Mean overall instructor teaching effectiveness rating: 4.6/5.0

Fall 2023 DS2000, Programming with Data (Undergraduate)
Enrollment: 278 (two sections)
Mean overall instructor teaching effectiveness rating: 4.3/5.0

Spring 2022 DS4420, Machine Learning 2 (Advanced Undergraduate)
Enrollment: 47
Mean overall instructor teaching effectiveness rating: 4.5/5.0

Fall 2021 DS2000, Programming with Data (Undergraduate)
Enrollment: 181 (two sections)
Mean overall instructor teaching effectiveness rating: 4.1/5.0

Spring 2021 DS4900, Senior Design Project (Advanced Undergraduate)
Enrollment: 22
Mean overall instructor teaching effectiveness rating: 4.8/5.0

Fall 2020 DS4440, Practical Neural Networks (Advanced Undergraduate)
Enrollment: 13
Mean overall instructor teaching effectiveness rating: 4.8/5.0

Spring 2020 DS4420, Machine Learning and Data Mining 2 (Advanced Undergraduate)
Enrollment: 16
Mean overall instructor teaching effectiveness rating: 4.7/5.0

Fall 2019 DS2000, Programming with Data (Undergraduate)

Enrollment: 45
Mean overall instructor teaching effectiveness rating: 4.7/5.0

Fall 2019 CS4950, Computer Science Research Seminar (Undergraduate)
Enrollment: 18
Mean overall instructor teaching effectiveness rating: 5.0/5.0

Spring 2019 DS4440, Practical Neural Networks (Advanced Undergraduate)
Enrollment: 13
Mean overall instructor teaching effectiveness rating: 4.9/5.0

Fall 2018 DS2000, Programming with Data (Undergraduate)
Enrollment: 48
Mean overall instructor teaching effectiveness rating: 4.6/5.0

Fall 2017 CS2500, Fundamentals I (Undergraduate)
Enrollment: 40
Mean overall instructor teaching effectiveness rating: 4.4/5.0

Spring 2017 CS4100, Artificial Intelligence (Undergraduate)
Course website and syllabus: `http://www.byronwallace.com/ccis4100`
Enrollment: 25
Mean overall instructor teaching effectiveness rating: 5.0/5.0

Fall 2016 CS7180, Special Topics: Machine Learning in Health (Graduate)
Course website and syllabus: `http://www.byronwallace.com/ccis7180`
Enrollment: 6
Mean overall instructor teaching effectiveness rating: 5.0/5.0

UT Austin

Spring 2016 INF385T, Applied Data Mining (Advanced Undergraduate/Graduate)
Enrollment: 15
Mean overall instructor rating: 4.9/5.0

Fall 2015 INF385M, Database Management (Advanced Undergraduate/Graduate)
Enrollment: 17
Mean overall instructor rating: 4.9/5.0

Spring 2015 INF385T, Applied Data Mining (Advanced Undergraduate/Graduate)
Enrollment: 19
Mean overall instructor rating: 4.2/5.0

Fall 2014 INF385T, Applied Data Mining (Advanced Undergraduate/Graduate)
Enrollment: 19
Mean overall instructor rating: 3.9/5.0

Advising & thesis committees

Advising

Current Northeastern CS PhD students

9/2026 – *present* Michael Li; Co-advised with David Bau

9/2025 – *present* Zihao (Gavin) Yang

9/2025 – *present* Xilin (Rice) Wang

9/2023 – *present* Sheridan Feucht; Co-advised with David Bau

9/2022 – *present* Hiba Ahsan

9/2022 – *present* Millicent Li

Northeastern CS undergraduates

1/2023–6/2024 Jiuding Sun: Received an Honorable Mention for the CRA *Outstanding Undergraduate Researchers award on my nomination in 2024. Pursing a PhD at Stanford from Fall 2024.*

7/2022–12/2022 Stephanie Martinez

9/2017–9/2020 Eric Lehman: Received an Honorable Mention for the CRA *Outstanding Undergraduate Researchers award on my nomination in 2020. Finished a PhD at MIT in 2024.*

1/2017–5/2017 Derek Schuster

Ph.D. Dissertations Supervised

7/2026 **Alberto Mario Ceballos Arroyo**, Computer Science, Northeastern

Co-advised with Huaizu Jiang (Northeastern).

Thesis title: *Improving the detection of aneurysms in 3D radiology scans through anatomically-informed deep learning*

Committee: Jennifer Dy (Northeastern), Geoffrey Young (Harvard Medical), Maria Zuluaga (EURECOM)

Now at: Postdoctoral Researcher at Cornell

6/2026 **Sanjana Ramprasad**, Computer Science, Northeastern

Thesis title: *Rethinking Factuality Assessment: Characterizing and Evaluating Hallucinations Across Specialized Domains*

Committee: David Bau (Northeastern), Terra Blevins (Northeastern), Greg Durrett (NYU)

Now at: Postdoctoral Researcher at Stanford

6/2026 **Hye Sun Yun**, Computer Science, Northeastern

Co-advised with Tim Bickmore (Northeastern).

Thesis title: *LLMs for Consumer Health Information: Understanding Users and Models for Responsible Health AI*

Committee: Mai ElSherief (Northeastern), Emma Pierson (UC Berkeley)

Recipient of the *Khoury College PhD Teaching Award*

Now at: Assistant Professor (TT) of Data Science at Lafayette College

4/2026 **Chantal Shaib**, Computer Science, Northeastern

Thesis title: *What Makes Large Language Model Text Distinct? Lexical and Syntactic Markers of Style*

Committee: David Bau (Northeastern), Malihe Alikhani (Northeastern), Adina Williams (Meta FAIR)

Recipient of the *Outstanding PhD Research Award*

Now at: Postdoctoral Researcher at MIT

11/2025 **Somin Wadhwa**, Computer Science, Northeastern

Thesis title: *From Large to Small: Distillation in the Age of Large Language Models*

Committee: David Bau (Northeastern), Silvio Amir (Northeastern), Hoifung Poon (Microsoft Research))

Now at: Applied Scientist in Core AI at Qualtrics

12/2024 **Jay DeYoung**, Computer Science, Northeastern

Thesis title: *Automation Assistance For Systematic Reviewers*

Committee: David Bau (Northeastern), Silvio Amir (Northeastern), Lucy Lu Wang (University of Washington)

Now at: Young Investigator at the Allen Institute for AI

3/2024 **Denis Jered McInerney**, Computer Science, Northeastern

Thesis title: *An Interface for Clinicians: Finding Crucial Information with Language Models in Electronic Health Records*

Committee: Silvio Amir (Northeastern), Jan-Willem van de Meent (Northeastern/University of Amsterdam), Marzyeh Ghassemi (MIT)

Now at: Research Scientist at CodaMetrix

7/2022 **Sarthak Jain**, Computer Science, Northeastern

Thesis title: *The Model thinks What?! Interpreting Deep NLP models with Rationales and Influence*

Committee: David Smith (Northeastern), Jan-Willem van de Meent (Northeastern/University of Amsterdam), Sameer Singh (University of California, Irvine)

Now at: Machine Learning Scientist at Profluent

8/2022 **Ben Nye**, Computer Science, Northeastern

Thesis title: *What Does the Evidence Say? Automatic Information Extraction from Medical Trial Reports*

Committee: David Smith (Northeastern), Mirek Riedewald (Northeastern), Ani Nenkova (UPenn/Adobe Research)

Recipient of the *PhD Teaching Award*

Now at: Assistant Professor in Computer Science (TT) at Colorado College

5/2019 **Ye Zhang**, Computer Science, University of Texas at Austin

Thesis title: *Neural NLP Models Under Low-Supervision Scenarios*

Co-advised with Matthew Lease (UT Austin).

Committee: Greg Durrett (UT Austin) and Raymond Mooney (UT Austin)

Now at: Google Brain

5/2019 **An Nguyen**, Computer Science, University of Texas at Austin
Thesis title: *Probabilistic Modeling with Human Factors in Machine Learning*
Co-advised with Matthew Lease (UT Austin).
Committee: Greg Durrett (UT Austin) and Qiang Liu (UT Austin)

PhD committees

2026 **Monica Munnangi**, Northeastern
Thesis title: *Towards Reliable Clinical AI: Evaluating and Enhancing the Performance of Large Language Models in Clinical Settings*
Committee: Saiph Savage (*Chair*; Northeastern), Malihe Alikhani (Northeastern), Aakanksha Naik (Allen Institute for AI)

2024 **Eric Lehman**, CSAIL, MIT
Thesis title: *Practical Considerations For the Deployment of Clinical NLP Systems*
Committee: Peter Szolovits (*Chair*; MIT), Marzyeh Ghassemi (MIT)

2024 **Kundan Krishna**, LTI, Carnegie Mellon University
Thesis title: *Improving the reliability of language models for summarization*
Committee: Zachary Lipton (*Co-Chair*; CMU), Jeffrey Bigham (*Co-Chair*; CMU), Sherry Tonghuang Wu (CMU), Alexander Rush (Cornell)

2024 **Wojciech Kusa**, TU Wien
Thesis title: *Automated Eligibility Screening and its Evaluation in the Medical Domain*
Committee: Alan Hanbury (*Chair*; TU Wien), Petr Knoth (TU Wien), Evangelos Kaoulas (University of Amsterdam)

2024 **Griffin Adams**, Computer Science and DBMI, Columbia University
Thesis title: *Generating Faithful and Complete Hospital-Course Summaries from the Electronic Health Record*
Committee: Smaranda Muresan (*Moderator*, Barnard College), Noémie Elhadad (*Co-Advisor*, Columbia), Kathleen McKeown (*Co-Advisor*, Columbia), Jason Zucker (Columbia)

2022 **Diego Garcia-Olano**, Electrical and Computer Engineering, University of Texas at Austin
Thesis title: *In-process Diagnostic methods for Entity Representation Learningon Sequential data at Scale*
Committee: Joydeep Ghosh (*Chair*; UT Austin), Vikalo Harris (UT Austin), Alexandros Dimakis (UT Austin), Atlas Wang (UT Austin)

2022 **Liwen Hou**, Khoury College of Computer Sciences, Northeastern University
Thesis title: *Detecting and Modeling Syntactic Change*
Committee: David Smith (*Chair*; Northeastern), Tina Eliassi-Rad (Northeastern), Jacob Eisenstein (Google AI)

2021 **Rui Dong**, Khoury College of Computer Sciences, Northeastern University
Thesis title: *Natural Language Processing on Noisy Text*

Committee: David Smith (*Chair*; Northeastern), Mirek Riedewald (Northeastern), Steven Bedrick (OHSU)

2021 **Olga Kovaleva**, Computer Science, UMass Lowell

Thesis title: *Transformer Models in Natural Language Understanding: Strengths, Weaknesses and Limitations*

Committee: Anna Rumshisky (*Chair*; UMass Lowell), Hong Yu (UMass Lowell), Tingjian Ge (UMass Lowell)

2021 **Stefan Olafsson**, Khoury College of Computer Sciences, Northeastern University

Thesis title: *A Hybrid Structured-Neural Dialog System for Effective Automated Counseling*

Committee: Tim Bickmore (*Chair*; Northeastern), Paola Pedrelli (Harvard Medical)

2021 **Abhyuday Jagannatha**, College of Computer and Information Science, University of Massachusetts at Amherst

Thesis title: *Machine Learning methods for Extracting Adverse Drug Interactions and Pharmacovigilance*

Committee: Hong Yu (*Chair*; UMass Lowell), Phil Thomas (UMass Amherst), Dan Sheldon (UMass Amherst)

2021 **Ansel MacLaughlin**, Khoury College of Computer Sciences, Northeastern University

Thesis title: *Analyzing the Usage of Source Texts in News Documents*

Committee: David Smith (*Chair*; Northeastern), David Lazer (Northeastern), John Wilby (Northeastern), Mark Dredze (JHU)

2021 **Xinyu Hua**, Khoury College of Computer Sciences, Northeastern University

Thesis title: *Improving Controllabibility for Neural Text Generation*

Committee: Lu Wang (*Chair*; Northeastern), David Smith (Northeastern), Vincent Ng (UT Dallas)

2021 **Harrisen Scells**, School of Information Technology and Electrical Engineering, University of Queensland

Thesis title: *Improving Systematic Review Creation With Information Retrieval*

Committee: Guido Zuccon (*Chair*; University of Queensland), Sun Aixin (Nanyang Technological University)

2020 **Xiaolei Huang**, Information Science, University of Colorado Boulder

Thesis title: *Metadata Matters: Personalizing Document Classifiers via Domain Adaptation*

Committee: Michael J. Paul (*Chair*; UC Boulder), Mark Dredze (JHU), James Martin (UC Boulder), Robin Burke (UC Boulder))

2019 **Shoabin Xu**, Khoury College of Computer Sciences, Northeastern

Thesis title: *Modeling Text Embedded Information Cascades*

Committee: David Smith (*Chair*; Northeastern), Tina Eliassi-Rad (Northeastern), Bruce Desmarais (Penn State)

2018 **Jesse Anderton**, Khoury College of Computer Sciences, Northeastern

Thesis title: *Scalable Ordinal Embedding to Model Text Similarity*

Committee: Javed Aslam (*Chair*; Northeastern), Fernando Diaz (NYU), David Smith (Northeastern)

2018 **Maryam Aziz**, Khoury College of Computer Sciences, Northeastern

Thesis title: *Pure Exploration for the Infinitely-Armed Bandit Models in Fixed-Confidence and Fixed-Budget settings*

Committee: Javed Aslam (*Chair*; Northeastern), Emilie Kaufmann (INRIA), Jonathan Ulman (Northeastern)

2017 **David Inouye**, Computer Science, University of Texas at Austin

Thesis title: *Appropriate, Accessible and Appealing Probabilistic Graphical Models*

Committee: Pradeep Ravikumar (*Chair*; Carnegie Mellon University), Raymond Mooney (UT Austin), Qixing Huang (UT Austin)

2016 **David Batista**, Computer Science, Universidade de Lisboa/INESC-ID

Thesis title: *Large-Scale Semantic Relationship Extraction for Information Discovery*

Committee: Mario Gaspar da Silva (*Chair*; Universidade de Lisboa/INESC-ID), Paulo Quaresma (Universidade de Évora), David Manuel Martins de Matos (Universidade de Lisboa)

2015 **Hyun Joon Jung**, School of Information, University of Texas at Austin

Thesis title: *Temporal Modeling of Crowd Work Quality for Quality Assurance in Crowdsourcing*

Committee: Matthew Lease (*Chair*; UT Austin), Paul Bennett (Microsoft Research), Ken Fleischmann (UT Austin), Raymond Mooney (UT Austin)

2015 **Louise Millard**, School of Social and Community Medicine, University of Bristol

Thesis title: *Towards data-intensive epidemiology: Explorations in systematic reviews and causal inference*

Committee: Julian Higgins (*Co-Chair*; University of Bristol), Peter Flach (*Co-Chair*; University of Bristol), James Thomas (UCL)

Service

Conference organization

NAACL I served as the *Local Sponsorship Chair* for NAACL 2022.

Semantic Scholar, Allen Institute for AI Advisory Board Member; 2023 – 2025.

MLHC Co-chair of the Machine Learning in Healthcare *Machine Learning in Healthcare* (MLHC) conference, 2016, 2017 (hosted locally at Northeastern), 2018 (Stanford), 2019 (UMich), and 2020 (remote due to Covid-19). From 2021, I have transitioned to the Board of Directors for MLHC.

COVID-19 workshop Co-organizer of the COVID-19 Workshop co-located at the Association for Computational Linguistics (ACL) conference, 2020.

NLPMC Co-organizer of NLP for Medical Conversations co-located at the Association for Computational Linguistics (ACL) conference, 2020.

W3PHI Co-organizer of The International Workshop on the World Wide Web and Public Health Intelligence (W3PHI) – a workshop co-located with AAAI in years 2016–2020 (inclusive).

MAIHA I co-chaired the workshop *Modern Artificial Intelligence in Health Analytics (MAIHA)* at AAAI 2014, along with Jenna Wiens, David Kale and Finale Doshi-Velez.

CogSci Workshop — I co-organized the workshop Can Cognitive Scientists Help Computers Recognize Irony? at CogSci 2014, in collaboration with Laura Kertz.

Grant & proposal reviewing

Review panel service — National Institutes of Health (NIH) in academic years 2015-2016, 2016-2017, 2017-2018, 2020-2021, and 2022-2023; also served on a panel for the National Library of Medicine (under the NIH) in 2018-2019. National Science Foundation (NSF) in academic years 2013-2014, 2014-2015, 2017-2018, 2019-2020, 2020-2021, 2021-2022, and 2022-2023.

Ad-hoc review — I have reviewed proposals remotely for the National Science Foundation (NSF) in academic year 2018, and for the Army Research Office (ARO), mathematical sciences division in academic year 2014-2015.

National Academies of Science

Workshop Committee — I served on a National Academies Workshop Committee to Support EPA's Development of Human Health Effects Assessments in 2022.

Peer reviewing &etc.

Action Editor — Transactions of the Association for Computational Linguistics (TACL); 2023 – present

Action Editor — ACL Rolling Review; 2022 – present

Action Editor — Computational Linguistics Journal; 2020 – 2024.

Action Editor — Machine Learning Journal (MLJ); 2016 – 2020.

Editorial Board Membership — I am on the editorial board of Research Synthesis Methods (RSM), and served as the *Computational Tools and Methods Editor* from 2017 to 2020.

Standing reviewer — TACL; 2018 – 2023

Guest editor — I guest edited a special issue in the Machine Learning Journal (MLJ) on applications in health (alongside Jenna Wiens).

(Senior) Area Chair Roles — I am a standing Area Chair (AC) for the *Association for Computational Linguistics* ACL Rolling Review (ARR). I was a Senior AC for *Annual Conference of the Association for Computational Linguistics* 2024 and in 2026, and an AC in 2023. I was a Senior AC for *Empirical Methods in Natural Language Processing (EMNLP)* in 2026 and was an AC for 2019–2022, and in 2024. I was an Senior AC for *Annual Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)* in 2021 and again in 2025, and served as an AC in 2016. I was an AC for the *Conference on Language Models (COLM)* 2024, 2025, and 2026. I was an AC for International Conference Machine Learning (ICML) 2025 – 2026, and a Senior AC for 2027. I was also an SPC member for *Knowledge Discovery and Data Mining (KDD)* in 2015. I served as an AC for *Neural Information Processing Systems (NeurIPS)* 2018 and 2019, and in 2022 and 2024, 2025, and 2026. I served as Senior Area Chair the for *Association for the Advancement of Artificial Intelligence (AAAI)* 2020—2022, and I served as a senior program committee (SPC) member for the same in 2018 and 2014. I served on the SPC for the International Joint Conference on Artificial Intelligence (IJCAI) 2020. I was an AC for the *International Conference on Learning Representations* (ICLR) 2021 – 2022, and 2024 – 2025 and a Senior AC for 2027.

Program committee memberships

2019 — International Conference on Machine Learning (ICML); International Conference on Learning Representations (ICLR); Association for Computational Linguistics (ACL); American Medical Informatics Association (AMIA) Joint Summits

2018 International Conference Machine Learning (ICML); International Joint Conference on Artificial Intelligence (IJCAI); International Conference on Learning Representations (ICLR); Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)

2017 Association for the Advancement of Artificial Intelligence (AAAI); Association for Computational Linguistics (ACL); European Association for Computational Linguistics (EACL); International Conference Machine Learning (ICML); Neural Information Processing (NIPS); International Joint Conference on Artificial Intelligence (IJCAI); World Wide Web conference (WWW)

2016 Association for the Advancement of Artificial Intelligence (AAAI); Association for Computational Linguistics (ACL); Neural Information Processing (NIPS); International Conference on Machine Learning (ICML); Knowledge Discovery and Data Mining (KDD); AAAI Conference on Human Computation and Crowdsourcing (HCOMP); International Joint Conference on Artificial Intelligence (IJCAI)

2015 Association for the Advancement of Artificial Intelligence (AAAI); American Medical Informatics Association (AMIA); European Conference on Machine Learning (ECML); International Conference on Machine Learning (ICML); Neural Information Processing (NIPS)

2014 American Medical Informatics Association (AMIA); European Conference on Machine Learning (ECML); International Conference on Data Mining (ICDM); Knowledge Discovery and Databases (SIGKDD) Workshop on Health Informatics)

2013 Advancement of Artificial Intelligence (AAAI); American Medical Informatics Association (AMIA); IEEE Conference on Big Data (ICBD)

2012 International Conference on Information and Knowledge Management (CIKM); Knowledge Discovery and Databases (SIGKDD) Workshop on Health Informatics

Journals for which I have reviewed

British Medical Journal (BMJ), BMC Bioinformatics, BMC Medical Informatics and Decision Making, BMC Medical Research Methodology, BMC Research Notes, Bioinformatics, Current Bioinformatics, Data Mining and Knowledge Discovery, IEEE Transactions on Knowledge and Data Engineering, International Journal of Epidemiology, Journal of the American Medical Association (JAMA), Journal of the American Medical Informatics Association (JAMIA), Journal of Biomedical Informatics (JBI), Journal of Clinical Epidemiology (JCE), Journal of Machine Learning Research (JMLR), Journal of Medical Internet Research (JMIR), Journal of Systematic Reviews, Knowledge and Information Systems (KAIS), Language Resources and Evaluation (LREC), Research Synthesis Methods (RSM)